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Medical device including attachable components

Abstract

An example system for delivering an implantable heart valve is disclosed. The system includes a connection assembly including: an inner shaft having a proximal end region, a distal end region and a first coupling member disposed along a portion of the distal end region, wherein the first coupling member includes a plurality of apertures. The connection assembly also includes a support shaft and a second coupling member, wherein the second coupling member includes a stem, wherein the stem includes a groove extending circumferentially around the stem. The connection assembly also includes a plurality of bearings, wherein a portion of each bearing is disposed in a portion of the groove. The connection assembly also includes a locking cap coupled to the inner shaft, wherein a portion of the locking cap is configured to cover at least a portion of each of the plurality of bearings.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit of priority of U.S. Provisional Application No. 63/141,795 filed Jan. 26, 2021, the entire disclosure of which is hereby incorporated by reference.

TECHNICAL FIELD

(1) The present disclosure pertains to medical devices, and methods for manufacturing medical devices. More particularly, the present disclosure pertains to medical devices including an attachable inner member and attachable outer member.

BACKGROUND

(2) A wide variety of intracorporeal medical devices have been developed for medical use, for example, intravascular use. Some of these devices include heart valves, catheters, and the like. These devices are manufactured by any one of a variety of different manufacturing methods and may be used according to any one of a variety of methods. Of the known medical devices and methods, each has certain advantages and disadvantages. There is an ongoing need to provide alternative medical devices as well as alternative methods for manufacturing and using medical devices.

BRIEF SUMMARY

(3) This disclosure provides design, material, manufacturing method, and use alternatives for medical devices. An example system for delivering an implantable heart valve includes a

connection assembly having an engaged configuration and a disengaged configuration, the connection assembly including: an inner shaft having a proximal end region, a distal end region and a first coupling member disposed along a portion of the distal end region, wherein the first coupling member includes a plurality of apertures. The connection assembly also includes a support shaft having a proximal end region, a distal end region and a second coupling member disposed along a portion of the proximal end region, wherein the second coupling member includes a stem, wherein the stem includes a groove extending circumferentially around the stem. The connection assembly also includes a plurality of bearings, wherein each bearing is disposed in a respective aperture of the plurality of apertures, and wherein a portion of each bearing is also disposed in a portion of the groove when the connection assembly is in the engaged configuration. The connection assembly also includes a locking cap coupled to the inner shaft, wherein a portion of the locking cap is configured to cover at least a portion of each of the plurality of bearings when the connection assembly is in the engaged configuration.

(4) Alternatively or additionally to any of the embodiments above, wherein the locking cap includes a plurality of tabs extending circumferentially around a longitudinal axis of the locking cap.

(5) Alternatively or additionally to any of the embodiments above, wherein each of the plurality of tabs is configured to flex radially away from the longitudinal axis.

(6) Alternatively or additionally to any of the embodiments above, wherein each of the plurality of tabs is configured to flex radially away from the longitudinal axis as the stem is inserted into a lumen of the first coupling member.

(7) Alternatively or additionally to any of the embodiments above, wherein each of the plurality of tabs is configured to exert a radially inward force on each of the plurality of bearings in the engaged configuration.

(8) Alternatively or additionally to any of the embodiments above, wherein each of the plurality of bearings is configured to slide circumferentially around the stem while disposed within the groove.

(9) Alternatively or additionally to any of the embodiments above, wherein each of the bearings is configured to rotate relative to the first coupling member, the second coupling member and the locking cap in the engaged configuration.

(10) Alternatively or additionally to any of the embodiments above, wherein each of the bearings is configured to shift radially outward within its respective aperture of the plurality of apertures as the stem is inserted into a lumen of the first coupling member.

(11) Alternatively or additionally to any of the embodiments above, wherein each of the plurality of bearings is disposed within its respective aperture of the plurality of apertures in the disengaged configuration.

(12) Alternatively or additionally to any of the embodiments above, wherein the locking cap is fixedly attached to the first actuation coupling member.

(13) Alternatively or additionally to any of the embodiments above, wherein the first coupling member is configured to rotate relative to the second coupling member in the engaged configuration.

(14) Another system for delivering an implantable heart valve includes a tip assembly having a distal end region and a proximal end region, a guidewire shaft coupled to the distal end region of the tip assembly and a connection assembly having an engaged configuration and a disengaged configuration, the connection assembly including: an inner shaft having a proximal end region, a distal end region and a first coupling member disposed along a portion of the distal end region, wherein the first coupling member includes a plurality of apertures. The connection assembly also includes a support shaft having a proximal end region, a distal end region and a second coupling member disposed along a portion of the proximal end region, wherein the second coupling member includes a stem, wherein the stem includes a groove extending circumferentially around the stem. The connection assembly also includes a plurality of bearings, wherein each bearing is disposed in

a respective aperture of the plurality of apertures, and wherein a portion of each bearing is also disposed in a portion of the groove when the connection assembly is in the engaged configuration. The connection assembly also includes a locking cap coupled to the inner shaft, wherein a portion of the locking cap is configured to cover at least a portion of each of the plurality of bearings when the connection assembly is in the engaged configuration.

(15) Alternatively or additionally to any of the embodiments above, wherein the locking cap includes a plurality of tabs extending circumferentially around a longitudinal axis of the locking cap.

(16) Alternatively or additionally to any of the embodiments above, wherein each of the plurality of tabs is configured to flex radially away from the longitudinal axis.

(17) Alternatively or additionally to any of the embodiments above, wherein each of the plurality of tabs is configured to flex radially away from the longitudinal axis as the stem is inserted into a lumen of the first coupling member.

(18) Alternatively or additionally to any of the embodiments above, wherein each of the plurality of tabs is configured to exert a radially inward force on each of the plurality of bearings in the engaged configuration.

(19) Alternatively or additionally to any of the embodiments above, wherein each of the plurality of bearings is configured to slide circumferentially around the stem while disposed within the groove.

(20) Alternatively or additionally to any of the embodiments above, wherein each of the bearings is configured to rotate relative to the first coupling member, the second coupling member and the locking cap in the engaged configuration.

(21) Alternatively or additionally to any of the embodiments above, wherein the first coupling member is configured to rotate relative to the second coupling member in the engaged configuration.

(22) An example method for delivering an implantable heart valve includes attaching a first coupling member of an actuation shaft to a second coupling member of a support shaft of a medical device delivery system, the medical device delivery system including the implantable heart valve, wherein attaching the first coupling member of the actuation shaft to the second coupling member of the support shaft includes positioning a plurality of bearings into a groove of the second coupling member. The method also includes advancing the medical device delivery system to a target site adjacent the heart; and deploying the implantable heart valve at the target site.

(23) The above summary of some embodiments is not intended to describe each disclosed embodiment or every implementation of the present disclosure. The Figures, and Detailed Description, which follow, more particularly exemplify these embodiments.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The disclosure may be more completely understood in consideration of the following detailed description in connection with the accompanying drawings, in which:

(2) FIG. 1 is a side view of an example medical device system;

(3) FIG. 2 is a side view of the tip assembly and valve assembly spaced away from the inner shaft and exoskeleton of the medical device of FIG. 1;

(4) FIG. 3 is a perspective view of a connection assembly of the medical device of FIG. 1;

(5) FIG. 4 is an exploded view of the connection assembly shown in FIG. 3;

(6) FIG. 5 is a cross-sectional view taken along line 5-5 of FIG. 3;

(7) FIG. 6 is a perspective view of an example component of the connection assembly shown in FIGS. 3-4;

(8) FIG. 7 is a perspective view showing multiple translation members and a securement collar of

the connection assembly shown in FIGS. 3-4;

(9) FIG. 8 is a cross-sectional view of the connection assembly in a disengaged configuration;

(10) FIG. 9 is a cross-sectional view of the connection assembly in an engaged configuration.

(11) While the disclosure is amenable to various modifications and alternative forms, specifics thereof have been shown by way of example in the drawings and will be described in detail. It should be understood, however, that the intention is not to limit the disclosure to the particular embodiments described. On the contrary, the intention is to cover all modifications, equivalents, and alternatives falling within the spirit and scope of the disclosure.

DETAILED DESCRIPTION

(12) For the following defined terms, these definitions shall be applied, unless a different definition is given in the claims or elsewhere in this specification.

(13) All numeric values are herein assumed to be modified by the term “about”, whether or not explicitly indicated. The term “about” generally refers to a range of numbers that one of skill in the art would consider equivalent to the recited value (e.g., having the same function or result). In many instances, the terms “about” may include numbers that are rounded to the nearest significant figure.

(14) The recitation of numerical ranges by endpoints includes all numbers within that range (e.g. 1 to 5 includes 1, 1.5, 2, 2.75, 3, 3.80, 4, and 5).

(15) As used in this specification and the appended claims, the singular forms “a”, “an”, and “the” include plural referents unless the content clearly dictates otherwise. As used in this specification and the appended claims, the term “or” is generally employed in its sense including “and/or” unless the content clearly dictates otherwise.

(16) It is noted that references in the specification to “an embodiment”, “some embodiments”, “other embodiments”, etc., indicate that the embodiment described may include one or more particular features, structures, and/or characteristics. However, such recitations do not necessarily mean that all embodiments include the particular features, structures, and/or characteristics. Additionally, when particular features, structures, and/or characteristics are described in connection with one embodiment, it should be understood that such features, structures, and/or characteristics may also be used connection with other embodiments whether or not explicitly described unless clearly stated to the contrary.

(17) The following detailed description should be read with reference to the drawings in which similar elements in different drawings are numbered the same. The drawings, which are not necessarily to scale, depict illustrative embodiments and are not intended to limit the scope of the invention.

(18) Diseases and/or medical conditions that impact the cardiovascular system are prevalent throughout the world. Traditionally, treatment of the cardiovascular system was often conducted by directly accessing the impacted part of the body. For example, treatment of a blockage in one or more of the coronary arteries was traditionally treated using coronary artery bypass surgery. As can be readily appreciated, such therapies are rather invasive to the patient and require significant recovery times and/or treatments. More recently, less invasive therapies have been developed. For example, therapies have been developed which allow a blocked coronary artery to be accessed and treated via a percutaneous catheter (e.g., angioplasty). Such therapies have gained wide acceptance among patients and clinicians.

(19) Some relatively common medical conditions may include or be the result of inefficiency, ineffectiveness, or complete failure of one or more of the valves within the heart. For example, failure of the aortic valve or the mitral valve can have a serious effect on a human and could lead to serious health condition and/or death if not dealt with properly. Treatment of defective heart valves poses other challenges in that the treatment often requires the repair or outright replacement of the defective valve. Such therapies may be highly invasive to the patient. Disclosed herein are medical devices that may be used for delivering a medical device to a portion of the cardiovascular system

in order to diagnose, treat, and/or repair the system. At least some of the medical devices disclosed herein may be used to deliver and implant a replacement heart valve (e.g., a replacement aortic valve, replacement mitral valve, etc.). In addition, the devices disclosed herein may deliver the replacement heart valve percutaneously and, thus, may be much less invasive to the patient. The devices disclosed herein may also provide a number of additional desirable features and benefits as described in more detail below.

(20) The figures illustrate selected components and/or arrangements of a medical device system **10**, shown schematically in FIG. **1**, for example. It should be noted that in any given figure, some features of the medical device system **10** may not be shown, or may be shown schematically, for simplicity. Additional details regarding some of the components of the medical device system **10** may be illustrated in other figures in greater detail. A medical device system **10** may be used to deliver and/or deploy a variety of medical devices to a number of locations within the anatomy. In at least some embodiments, the medical device system **10** may include a replacement heart valve delivery system (e.g., a replacement aortic valve delivery system) that can be used for percutaneous delivery of a medical implant **16** (shown in the detailed view of FIG. **1**), such as a replacement/prosthetic heart valve. This, however, is not intended to be limiting as the medical device system **10** may also be used for other interventions including valve repair, valvuloplasty, delivery of an implantable medical device (e.g., such as a stent, graft, etc.), and the like, or other similar interventions.

(21) The medical device system **10** may generally be described as a catheter system that includes an outer shaft **12**, an exoskeleton **14** extending at least partially through a lumen of the outer shaft **12**, and a medical implant **16** (e.g., a replacement heart valve implant) which may be coupled to the exoskeleton **14** and disposed within a lumen of the outer shaft **12** during delivery of the medical implant **16**. In some embodiments, a medical device handle **18** may be disposed at a proximal end of the outer shaft **12** and/or the exoskeleton **14** and may include one or more actuation mechanisms associated therewith. In other words, one or more tubular members (e.g., the outer shaft **12**, the exoskeleton **14**, etc.) may extend distally from the medical device handle **18**. In general, the medical device handle **18** may be designed to manipulate the position of the outer shaft **12** relative to the exoskeleton **14** and/or facilitate the deployment of the medical implant **16**.

(22) In use, the medical device system **10** may be advanced percutaneously through the vasculature to a position adjacent to an area of interest and/or a treatment location. For example, in some embodiments, the medical device system **10** may be advanced through the vasculature to a position adjacent to a defective native valve (e.g., aortic valve, mitral valve, etc.). Alternative approaches to treat a defective aortic valve and/or other heart valve(s) are also contemplated with the medical device system **10**. During delivery, the medical implant **16** may be generally disposed in an elongated and low profile “delivery” configuration within the lumen and/or a distal end of the outer shaft **12**, as seen schematically in FIG. **1**, for example. Once positioned, the outer shaft **12** may be retracted relative to the medical implant **16** and/or the exoskeleton **14** to expose the medical implant **16**. In some instances, the medical implant **16** may be self-expanding such that exposure of the medical implant **16** may deploy the medical implant **16**. Alternatively, the medical implant **16** may be expanded/deployed using the medical device handle **18** in order to translate the medical implant **16** into a generally shortened and larger profile “deployed” configuration suitable for implantation within the anatomy. When the medical implant **16** is suitably deployed within the anatomy, the medical device system **10** may be disconnected, detached, and/or released from the medical implant **16** and the medical device system **10** can be removed from the vasculature, leaving the medical implant **16** in place in a “released” configuration.

(23) It can be appreciated that during delivery and/or deployment of an implantable medical device (e.g., the medical implant **16**), portions of the medical device system (e.g., the medical device system **10**) may be required to be advanced through tortuous and/or narrow body lumens. Therefore, it may be desirable to utilize components and design medical delivery systems (e.g.,

such as the medical device system **10** and/or other medical devices) that reduce the profile of portions of the medical device while maintaining sufficient strength (compressive, torsional, etc.) and flexibility of the system as a whole.

(24) In some instances, it may be desirable to design the medical device system **10** such that one or more device components may be disconnected from the medical device handle **18** when initially packaged (e.g., unattached to the exoskeleton **14**, other inner shafts, etc.) whereby the one or more components may be subsequently coupled to the handle **18** after the packaging containing the medical device system **10** has been opened (and prior to a clinician utilizing the medical device system **10** in a medical procedure). For example, in some instances it may be desirable to package the medical implant **16** (e.g., heart valve, heart valve frame, the heart valve support structure, etc.) separately prior to performing the medical procedure. It can be appreciated that packaging the medical implant **16** (e.g., heart valve, heart valve frame, the heart valve support structure, etc.) separately may permit the medical implant **16** (including the heart valve, heart valve frame, the heart valve support structure, etc.) to be sterilized according to a different process, or kept at different temperatures, for example, than the remaining separately-packaged components of the medical device system **10**.

(25) FIG. 2 shows an illustration of the medical device system **10** whereby the medical implant **16**, the medical implant support structure **26** (coupled to the medical implant **16**) and the tip assembly **24** are uncoupled from the handle **18** (it is noted that, for simplicity, the handle **18** is not shown in FIG. 2). It can be appreciated from FIG. 2 that any one of the medical implant **16**, the medical implant support structure **26** and/or the tip assembly **24** may be packaged separately from the remaining components (e.g., handle **18**, outer shaft **12**, exoskeleton **14**, guidewire shaft **36**, etc.) of the medical device system **10**, as described above.

(26) As discussed above, FIG. 2 illustrates that the tip assembly **24** is uncoupled (e.g., unattached) from the medical implant **16**, the medical implant support structure **26** and the remainder of the medical device delivery system **10**. For example, in the packaging of the medical device system **10**, the tip assembly may be packaged separately from the remainder of the medical device system **10**. However, FIG. 2 further illustrates that the tip assembly **24** may eventually be coupled to the handle member **18** (and remainder of the medical device system **10**) via a tubular guidewire member **36** (as illustrated by the dotted line **45**).

(27) In some examples, the tubular guidewire member **36** may extend proximally within the lumen of an exoskeleton **14** and couple to the handle member **18** (it is noted that the exoskeleton **14** will be discussed in greater detail below). Additionally, the tubular guidewire member **36** may include a lumen which permits a guidewire to extend and translate therein. In other words, when fully assembled, the medical device system **10** may be advanced to a target site within a body over a guidewire extending within the lumen of the tubular guidewire member **36**. Further, as discussed above, the tubular guidewire member **36** may extend from the handle member **18**, through the lumen of the exoskeleton **14**, through the implant medical and terminate at the tip assembly **24**. Additionally, to attach the tubular guidewire member **36** to the tip assembly **24**, the tubular guidewire member **36** may be advanced through the medical implant support structure **26** and the medical implant **16**. Further, the tip assembly **24** and the tubular guidewire member **36** may be designed such that they “quick connect” (e.g., snap, attach, engage, etc.) together. Examples of attaching the tip assembly to a tubular guidewire member **36** are disclosed in U.S. Patent Application No. 62/887,088, the entirety of which is incorporated by reference.

(28) As discussed above, FIG. 2 further illustrates the medical implant **16** (e.g., a heart valve) coupled to a medical implant support structure **26**. FIG. 2 illustrates that the medical implant **16** and the medical implant support structure **26** are uncoupled (e.g., unattached) from the remainder of the medical device delivery system **10**. In the configuration shown, it can be appreciated that the medical implant support structure **26** may include one or more components and/or features which are designed to maintain the medical implant **16** in a pre-delivery configuration prior to attaching

the medical implant **16** and medical implant support structure **26** to the remainder of the medical device system **10**.

(29) While FIG. **2** illustrates the medical implant **16** and the medical implant support structure **26** unattached to the remainder of the medical device system **10**, it can be appreciated that the medical implant **16** and the medical implant support structure **26** may be coupled to the remainder of the medical device system **10** (e.g., handle **18**) via one or more shaft members and/or coupling members (as illustrated by the dotted line **49**). The coupling of the medical implant **16** and the medical implant support structure **26** to the medical device system **10** will be described below.

(30) For example, as discussed above, FIG. **2** illustrates that the medical device system **10** may include an exoskeleton **14** extending within the outer shaft **12**. The exoskeleton **14** may include one or more lumens extending therein. One or more inner shafts may extend through the exoskeleton **14**. For example, the exoskeleton **14** may include a lumen through which an actuation shaft **17** may extend (the actuation shaft **17** will be described in greater detail below).

(31) Further, in some examples, the exoskeleton **14** may include a plurality of discrete members or articulating links. For example, the exoskeleton **14** may include a plurality of bead members **41** and a plurality of barrel members **43**. Other discrete members are contemplated that may have differing shapes and/or configurations. In general, the discrete members (e.g., the bead members **41** and the barrel members **43**) are engaged with one another and are designed to increase the compression resistance, the tension resistance, or both of the exoskeleton **14** while also affording a desirable amount of flexibility and kink resistance such that the one or more inner shafts extending through the exoskeleton can be navigated through the anatomy. The bead members **41** and the barrel members **43** may be arranged in a number of different configurations. In at least some instances, the bead members **41** and the barrel members **43** alternate along the exoskeleton **14**. Other arrangements and/or patterns are contemplated. Example exoskeletons are disclosed in U.S. Patent Publication No. US20180140323, the entirety of which is incorporated by reference.

(32) Additionally, FIG. **2** illustrates that, in some examples, the distal end of the exoskeleton **14** may include a first exoskeleton coupling member **30**. As will be described in greater detail below, the first exoskeleton coupling member **30** may include one or more features which are designed to attach to a second exoskeleton coupling member **28**. As further illustrated in FIG. **2**, the second exoskeleton coupling member **28** may be attached to the proximal end of one or more components of the medical implant support structure **26**. Therefore, it can be appreciated that coupling the first exoskeleton coupling member **30** to the second exoskeleton coupling member **28** may connect the exoskeleton **14** to the medical implant **16** via the medical implant support structure **26**.

(33) Additionally, as will be described in greater detail below, FIG. **2** illustrates that the medical device system **10** may include an exoskeleton locking collar **34**. The exoskeleton locking collar **34** may be disposed along an outer surface of the exoskeleton **14**. As will be described in greater detail below, the exoskeleton locking collar **34** may be utilized to couple (e.g., attach, lock, engage, etc.) the first exoskeleton coupling member **30** to the second exoskeleton coupling member **28**.

(34) It is noted that FIG. **2** illustrates the outer shaft **12** of the medical device system **10** having been retracted in a proximal direction to a position proximal of both the first exoskeleton coupling member **30**, the exoskeleton locking collar **34**, a portion of the actuation shaft **17** and a portion of the tubular guidewire member **36**. It can be appreciated that when all the components of the medical device system **10** (including the medical implant **16**, the medical implant support structure **26** and the tip assembly **24**) are assembled, the outer shaft **12** may be advanced distally such that it covers the medical implant **16**, the medical implant support structure **26** and a portion of the tip assembly **24**.

(35) Additionally, as discussed above, FIG. **2** illustrates that the medical device system **10** may include an actuation shaft **17** extending within a portion of the exoskeleton **14**. FIG. **2** further illustrates that, in some examples, the distal end of the actuation shaft **17** may include a first actuation shaft coupling member **19**. As will be described in greater detail below, the first actuation

shaft coupling member **19** may include one or more features which are designed to attach to a second actuation shaft coupling member **20**. As further illustrated in FIG. 2, the second actuation coupling member **20** may be attached to the proximal end of one or more translation members **22** (e.g., push-pull members). Therefore, it can be appreciated that coupling the first actuation shaft coupling member **18** to the second actuation coupling member **20** may connect the actuation shaft **17** to the medical implant **16** via the one or more translation members **22** (as illustrated by the dotted line **47**).

(36) In some examples, an operator may be able to manipulate the translation members **22** via the handle **18** (which is coupled to the translation members **22** via the actuation shaft **17**, first actuation coupling member **19** and second actuation coupling member **20**). For example, the handle **18** may be designed to control the translation of the translation members **22**. Further, actuation of the translation members **22** may help deploy the medical implant **16** at a target site adjacent the heart. Example translation members are disclosed in U.S. patent application Ser. No. 16/396,089, the entirety of which is incorporated by reference.

(37) Additionally, as will be described in greater detail below, FIG. 2 illustrates that the medical device system **10** may include an actuation shaft locking cap **32**. The actuation shaft locking cap **32** may be disposed along an outer surface of the actuation shaft **17**. As will be described in greater detail below, the actuation shaft locking cap **32** may be utilized to couple (e.g., attach, lock, engage, etc.) the first actuation shaft coupling member **19** to the second actuation coupling member **20**.

(38) In some instances, the order of connecting separately packaged components may include first advancing the guidewire shaft **36** through the medical implant. Next, the first actuation coupling member **19** may be attached to the second actuation coupling member **20**. After this connection is made, the actuation shaft **17** may be retracted such that the first exoskeleton coupling member **30** may be attached to the implant support structure **26** via the second exoskeleton coupling member **28**. Finally, the nosecone **24** may be attached to the distal end region of the guidewire shaft **36**.

(39) FIG. 3 is a perspective view showing a connection assembly **15**. Specifically, FIG. 3 illustrates the first actuation coupling member **19** coupled to the second actuation coupling member **20**. As shown in FIG. 3, the proximal end of the first actuation coupling member **19** may be attached to the distal end of the actuation shaft **17**. It can be appreciated from FIG. 3 that the first actuation coupling member **19** may include a lumen (not visible in FIG. 3, but shown in FIG. 4) extending therein. It can be further appreciated from FIG. 3 that the distal end region of the second actuation coupling member **20** may include a stem (not visible in FIG. 3, but shown in FIG. 4) which is configured to be inserted into the lumen of the first actuation coupling member **19**. A more detailed description of the first actuation coupling member **19** coupled to the second actuation coupling member **20** is provided below.

(40) Additionally, FIG. 3 illustrates the actuation shaft locking cap **32** disposed along the outer surface of the first actuation coupling member **19**. In some examples, the actuation shaft locking cap **32** may be fixedly attached (e.g., welded) to the first actuation coupling member **19**. As will be described in greater detail below, the actuation shaft locking cap **32** may be designed such that both the actuation shaft **17** and the proximal end region of the first actuation coupling member **19** may extend through a portion of the actuation shaft locking cap **32** (e.g., a portion of the actuation shaft locking cap **32** may extend circumferentially around the outer surface of the first actuation coupling member **19**). Additionally, FIG. 3 illustrates that the actuation shaft locking cap **32** may include one or more tabs **32** circumferentially spaced around the longitudinal axis of the locking cap **32**. Further details of the actuation locking cap **32** and its engagement with both the first actuation coupling member **19** and the second actuation coupling member **20** are discussed below.

(41) FIG. 3 further illustrates that the proximal end of the second actuation coupling member **20** may be attached to the distal ends of each of the translation members **22** described above. For example, it can be appreciated from FIG. 3 that the proximal end of the second actuation coupling member **20** may include a lumen (not visible in FIG. 3) within which the distal end region of each

of the translation members **22** may be inserted. Additional details of the engagement of the second actuation coupling member **20** with the translation members **22** is described below.

(42) FIG. **4** illustrates an exploded view of the connection assembly **15** shown in FIG. **3**.

Specifically, FIG. **4** illustrates the translation members **22** (coupled to one another via a securement collar **54**) aligned with the second actuation coupling member **20**, the second actuation coupling member **20** aligned with the first actuation coupling member **19**, and the actuation shaft locking cap **32** positioned proximal of the first actuation coupling member **19** and disposed along the actuation shaft **17**.

(43) FIG. **4** illustrates the translation members **22** aligned with the lumen **48** of the second actuation coupling member **20**. It is noted that FIG. **4** illustrates three translation members **22** positioned adjacent to one another (however, one of the translation members **22** is partially obstructed from view as it is positioned behind two other translation members **22** in the illustration). FIG. **4** further illustrates that the translation members **22** may be secured to one another via a securement collar **54**. The securement collar **54** may extend around all or a portion of the outer surface of the translation members **22**. Details of the engagement of the securement collar **54** and the translation members **22** is discussed with respect to FIG. **7** below.

(44) FIG. **4** further illustrates the second actuation coupling member **20** having a proximal end region **51** and a distal end region **44**. Additionally, the second actuation coupling member **20** may include a lumen **48** extending within a portion of the proximal end region **51**. As discussed above, the lumen **48** of the second actuation coupling member **20** may be sized to permit the distal end regions (including the securement collar **54**) of the translation members **22** to be inserted therein. It can be further appreciated from FIG. **4** that the distal ends of the translation members **22** may include a tapered (e.g., chamfered, beveled, angled, etc.) distal end which may permit easier insertion of the distal ends of the translation members **22** into the lumen **48** of the second actuation coupling member **20**.

(45) In some instances, the translation members **22** may be secured to the second actuation coupling member **20** via welding the securement collar **54** directly to the second actuation coupling member **20**. Further, in some examples, the aperture **50** may be utilized during the welding process to assure that the securement collar **54** is properly aligned with the second actuation coupling member **20**.

(46) As discussed above, FIG. **4** illustrates that the distal end region **44** of the second actuation coupling member **20** may be referred to as a stem extending away from the proximal end region **51**. As shown in FIG. **4**, the stem **44** may include an outer diameter which is less than the diameter of the lumen **48** of the proximal end region **51**. Additionally, in some examples, the second actuation coupling member **20** may include a tapered lip **55** which is positioned between the proximal end region **51** and the stem **44**.

(47) FIG. **4** further illustrates that the stem **44** may include a groove **46** which extends circumferentially around the circumference of the stem **44**. The groove **46** may be described as a portion of the stem **44** which extends radially inward from an outer surface of the stem **44**. Further, FIG. **4** illustrates that the groove **46** may be positioned along the stem **44** such that it is positioned approximately midway along the length of the stem **44**.

(48) However, this is not intended to be limiting. The groove **46** may be positioned along any portion of the stem **44**. Additionally, FIG. **4** illustrates that the stem **44** may include a tapered (e.g., chamfered, beveled, angled, etc.) distal end **53** which may permit easier insertion of the distal end of the stem **44** into the lumen **52** of the first actuation coupling member **19**.

(49) FIG. **4** further illustrates the first actuation coupling member **19** aligned with the second actuation coupling member **20**. As discussed above, the first actuation coupling member **19** may include a lumen **52** which may be configured to accept the stem **44** of the second actuation coupling member **20** therein. In other words, the lumen **52** may include a diameter which is sized to permit the outer diameter of the stem **44** be inserted therein.

(50) FIG. 4 further illustrates that the first actuation coupling member **19** may include one or more apertures **42** which extend from the outer surface of the first actuation coupling member **19**, through the wall of the first actuation coupling member **19** to an inner surface of the first actuation coupling member **19**. In other words, each aperture **42** extends from the outer surface of the first actuation coupling member **19** and opens into the lumen **52** of the first actuation coupling member **19**. Each of the apertures **42** may be spaced away from one another around the circumference of the first actuation coupling member **19**. For example, the first actuation coupling member **19** may include 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 or more apertures **42**, whereby the apertures **42** may be evenly spaced (e.g., symmetrically spaced) around the circumference of the first actuation coupling member **19**. However, in other examples the apertures **42** may be unevenly spaced around the circumference of the first actuation coupling member **19**. FIG. 4 further illustrates the actuation shaft locking cap **32** disposed along the actuation shaft **17**.

(51) FIG. 4 further illustrates that one or more spherical bearings **56** may be positioned within each aperture **42**. The bearings **56** may be designed to freely spin while positioned within the aperture **42**. Additionally, as will be shown in FIG. 5 below, each bearing **56**, while positioned within its respective aperture **42**, may extend radially inward (through the aperture **42**) such that a portion of each bearing **56** may nest within the groove **46**, while also being prevented from “falling through” its respective aperture **42**. This feature will be discussed in greater detail with respect to FIG. 5 below.

(52) Further, it can be appreciated that each individual bearing **56** may be designed to rotate, slide, spin, translate both within the aperture **42** and while nested within the groove **46**. Additionally, the bearings **56** may be designed to rotate around the circumference of the stem **44** while remaining within both the apertures **42** and the groove **46**. In other words, each of the bearings **56** may be permitted to travel around the outer surface of the stem **44** within the groove **46**, and while doing so, each individual bearing **56** may be spinning, sliding, rotating, etc. Referring back to FIG. 3, a portion of the locking cap **32** may extend overtop of each of the spherical bearings **56**, and therefore, may prevent each bearing from “falling out” of the upper opening of the aperture **42**. For example, the locking cap **32** may include one or more locking tabs **38** which may extend overtop of each of the spherical bearings **56**.

(53) FIG. 5 illustrates a cross-sectional of the connection assembly **15** taken along line 5-5 of FIG. 3. FIG. 5 illustrates each of the spherical bearings **56** positioned within the aperture **42** of the first actuation coupling member **19**. Additionally, as described above, FIG. 5 illustrates a portion of each spherical bearing extending radially inward of the inner surface of the first actuation coupling member **19** (the inner surface of coupling member **19** defining the lumen **52** shown in FIG. 4) such that each spherical bearing **56** rests within the groove **46** of the stem **44**. It can be appreciated from FIG. 5 that the apertures **42** are designed to permit only a portion of each spherical bearing to extend into the groove **46** (e.g., the spherical bearings are prevented from falling through the apertures **42** and into the lumen **58** when the stem **44** is removed from the lumen **52**).

(54) FIG. 5 further illustrates the locking cap **32** being positioned overtop of both the first actuation coupling member **19** and each of the spherical bearings **56**. It can be appreciated that the locking cap **32** may be designed to prevent each of the spherical bearings **56** from falling out of the aperture in which it is positioned while also permitting the bearing to spinning, sliding, rotating, etc. within the aperture **42** and the groove **46**.

(55) FIG. 6 illustrates a detailed view of the actuation shaft locking cap **32**. As shown in FIG. 6, the actuation shaft locking cap **32** may include a first end **68**, a second end **65** opposite the first end **68**, and one or more locking tabs **38** extending along the longitudinal axis from the first end **68** to the second end **65**. As shown in FIG. 6, the locking tabs **38** may be arranged circumferentially around the longitudinal axis **70** of the locking cap **32**. In some examples, the locking cap **32** may include one or more gaps **40** extending along the longitudinal axis **70**, whereby each gap **40** separates two individual locking tabs **38**.

(56) It can be appreciated that FIG. 6 illustrates that each locking tab **38** may have a uniform arc length X (around the longitudinal axis **70**) and that each gap **40** may have a uniform arc width Y (around the longitudinal axis **70**), this is not intended to be limiting. Rather, it is contemplated that the one or more locking tabs **38** may include variable arc lengths (e.g., the arc lengths X of one or more locking tabs **38** may be different) and the one or more gaps **40** may include variable arc lengths (e.g., the arc lengths Y of one or more gaps **40** may be different). Additionally, FIG. 6 illustrates that each locking tab **38** may include a length L, extending proximally from the second end **65** of the locking cap **32**. In some examples, such as the example shown in FIG. 6, the length L of each locking tab **38** may be uniform, however, in other examples, the locking cap **32** may include locking tabs **38** having variable lengths L. In other words, in some examples, one or more of the locking tabs **38** may include tabs having variable lengths L.

(57) As will be described in greater detail below, each of the locking tabs **38** may be designed to exert a force radially inward, toward the longitudinal axis **70**. It can be appreciated that the thickness of each locking tab **38** may correspond the amount of radially force exerted by each individual locking tab **38**. For example, increasing the thickness of locking tabs **38** may result in the locking tabs **38** being more resistant to flexing radially outward, and therefore, may exert increased forces radially inward.

(58) However, each of the locking tabs **38** may also be designed such that they may flex radially outward such that they can be advanced over one or more portions of the first actuation coupling member **19** and/or the second actuation coupling member **20**. The inward biasing force and the flexural capabilities of the locking cap **32** will be discussed in greater detail with respect to FIG. 8 below.

(59) FIG. 7 illustrates a partial exploded view of the securement collar **54** spaced away from the distal end regions of the translation members **22**. As shown in FIG. 7, the distal end region of each of the translation members **22** may include a notched portion **60**. It can be appreciated from FIG. 7, that each of the notched portions **60** may be circumferentially offset from one another approximately 120 degrees. Additionally, FIG. 7 illustrates that the securement collar **54** may include an inner surface **59**. Further, the inner surface **59** may include multiple flat surfaces **58**, each of which may be offset 120 degrees from one another. It can be further appreciated that when the securement collar **54** is coupled to the translation members **22**, each of the flat surfaces **58** may mate with each of the notched portion **60**, respectively, of the securement collar **54**.

(60) It can be further appreciated that the securement collar **54** may include a gap **69** which prevents the securement collar **54** from extending continuously around its longitudinal axis **70** (it is noted that the securement collar **54** shares the same longitudinal axis **70** as the locking cap **32** described above). Therefore, the securement collar **54** may be designed to flex such that it may flex around and onto (e.g., it may snap onto) the translation members **22** (collectively) whereby each of the flat surfaces **58** engage each of the notched regions **60**, respectively.

(61) FIG. 8 illustrates a cross-sectional view of the stem **44** of the second actuation coupling member **20** being inserted into the lumen **52** (as depicted by the arrow **74**) of the first actuation coupling member **19**. FIG. 8 may be described as a “disengaged” configuration of the connection assembly **15** shown in FIG. 3 (e.g., disengaged prior to the connection assembly being fully engaged as shown in below in FIG. 9). It can be appreciated from FIG. 8 that as the stem **44** is inserted into the lumen **52**, the portion of the outer surface of the stem **44** distal to the groove **46** may engage each of the spherical bearings **56**, thereby pushing each of the spherical bearings **56** radially outward as the stem is advanced into the lumen **52**. As each of the spherical bearings **56** is pushed radially outward, they may push against the locking tabs **38** of the locking collar **32** (which is positioned overtop a portion of the first actuation coupling member **19**). However, FIG. 8 illustrates the locking tabs **38** of the locking cap **32** may be designed to flex radially outward while simultaneously maintaining an inward radially force such that each spherical bearing is permitted to shift radially outward to allow the stem **44** to be inserted into the lumen **52**, but not to an extent

in which the bearings **56** fall out of the apertures **42**. It can be permitted that the spherical bearings **56** may remain in this position (and the locking tabs **38** may remain flexed) until the stem **44** is advanced to a position in which each of the bearings falls radially inward into the groove **46**. (62) FIG. **9** illustrates a cross-section of the connection assembly **15** shown in FIG. **8** after the spherical bearings **56** have been seated into the groove **46** (e.g., FIG. **9** illustrates the “engaged” configuration of the connection assembly **15**). FIG. **9** may be described as an “engaged” configuration of the connection assembly **15** shown in FIG. **8**. As described above, FIG. **9** illustrates two translation members **22** (it can be appreciated that the third translation member **22** is hidden behind the two visible translation members **22** shown in FIG. **7**) inserted into the lumen **48** of the proximal end region **51** of the second actuation coupling member **20**. Further, FIG. **9** illustrates the securement collar **54** positioned within the notched portions **60** of the translation members **22**. As discussed above, the securement collar **54** may be welded directly to the second actuation coupling member **20**.

(63) FIG. **9** further illustrates the stem **44** of the second actuation coupling member **20** positioned within the lumen **52** of the first actuation coupling member **19**. FIG. **9** further illustrates a portion of the spherical bearings **56** positioned within the groove **46** of the stem **44**. FIG. **9** illustrates that the cross-sectional shape of the groove **46** may resemble a “half circle” having a radius of curvature which mirrors the radius of curvature of the groove **46**.

(64) As discussed above, each individual bearing **56** may rotate, slide, spin, translate, etc. around the circumference of the stem **44** while remaining within the groove **46**. In other words, each of the bearings **56** is permitted to travel around the outer surface of the stem **44** within the groove **46**, and each individual bearing **56** may be spinning, sliding, rotating, etc. while doing so. As described above, a portion of each individual spherical bearing **56** may be positioned in a single aperture **42**, whereby the portion of the bearing **56** which extends radially outward of the groove **46** (e.g., the portion of the bearing which is not nested within the groove **46**) may be positioned within the aperture **46**. Additionally, as described above, FIG. **9** illustrates that locking tabs **38** of the locking collar positioned over top each of the spherical bearings **56**, thereby preventing them from falling out of each of the apertures **42**.

(65) It can be further appreciated that when the bearings **56** are positioned within the groove **46**, the locking cap **32** may be prevented from moving distally relative to the first actuation coupling member **19**. In other words, the locking cap **32** may abut spherical bearings **56**, which, in turn, abut the wall defining the aperture **42** of the first actuation coupling member **19**.

(66) Further, the engagement of each of the spherical bearings **56** with both a portion of the first actuation coupling member **19** (e.g., the bearings **56** engage the wall defining the aperture **42** of the first actuation coupling member **19**) and a portion of the second actuation coupling member **20** (e.g., the bearings **56** are nested with the groove **46** of the stem **44** of the second actuation coupling member **20**), prevents the first actuation coupling member **19** from being disengaged from the second actuation coupling member **20**. In other words, after the stem **44** is inserted into the lumen **52** to a position in which the spherical bearings **56** fall into the groove **46**, the stem **44** is prevented from being pulled out of the lumen **52**.

(67) Additionally, the detailed view of FIG. **9** illustrates the spherical bearing **56** resting within the groove **46** while also being positioned within a portion of the aperture **42**. Additionally, the detailed view of FIG. **9** illustrates that, when the spherical bearing **56** is positioned within the aperture **42**, the spherical bearing **56** may contact the angled surface of the aperture **42**. Further, the detailed view illustrates a contact point **61** between the spherical bearing **56** and the angled surface of the aperture **42**. The detailed view further illustrates an arrow **63** which represents a force vector extending from the contact point **61** and passes through the center of the spherical bearing **56**. It can be appreciated that the angled surface of the aperture **42** must be great enough such that the force vector **63** passes below the outer surface **62** of the stem **44**. It can be further appreciated that if the angled surface of the aperture **42** is not steep enough, the inward force imparted from the wall

of the aperture **42** to the spherical bearing **56** and, subsequently to the groove **46**, may not be great enough to prevent the spherical bearing **56** from being dislodged from the groove **46** under tensile loading conditions.

(68) Additionally, it can be appreciated that when the spherical bearings **56** are nested within the groove **46**, the second actuation coupling member **20** may be permitted swivel (e.g., spin along its longitudinal axis) relative to both the first locking actuation member **19** and the actuation shaft locking cap **32**. Because the spherical bearings **56** may spin, rotate, etc. within the groove **46**, the first actuation coupling member **19** may swivel relative to the second actuation coupling member **20**.

(69) Some example materials that can be used for the various components of the medical device system **10** are described herein. However, this is not intended to limit the devices and methods described herein, as the other materials may be utilized for the medical device system **10** and components thereof.

(70) Additionally, medical device system **10** and components thereof may be made from a metal, metal alloy, polymer (some examples of which are disclosed below), a metal-polymer composite, ceramics, combinations thereof, and the like, or other suitable material. Some examples of suitable polymers may include polytetrafluoroethylene (PTFE), ethylene tetrafluoroethylene ethylene propylene (FEP), polyoxymethylene (POM, for example, DELRIN® available from DuPont), polyether block ester, polyurethane (for example, Polyurethane 85A), polypropylene (PP), polyvinylchloride (PVC), polyether-ester (for example, ARNITEL® available from DSM Engineering Plastics), ether or ester based copolymers (for example, butylene/poly(alkylene ether) phthalate and/or other polyester elastomers such as HYTREL® available from DuPont), polyamide (for example, DURETHAN® available from Bayer or CRISTAMID® available from Elf Atochem), elastomeric polyamides, block polyamide/ethers, polyether block amide (PEBA, for example available under the trade name PEBAX®), ethylene vinyl acetate copolymers (EVA), silicones, polyethylene (PE), high density polyethylene (HDPE), polyester, Marlex high-density polyethylene, Marlex low-density polyethylene, linear low density polyethylene (for example REXELL®), ultra-high molecular weight (UHMW) polyethylene, polypropylene, polybutylene terephthalate (PBT), polyethylene terephthalate (PET), polytrimethylene terephthalate, polyethylenec naphthalate (PEN), polyetheretherketone (PEEK), polyimide (PI), polyetherimide (PEI), polyphenylene sulfide (PPS), polyphenylene oxide (PPO), poly paraphenylene terephthalamide (for example, KEVLAR®), polysulfone, nylon, nylon-12 (such as GRILAMID® available from EMS American Grilon), perfluoro (propyl vinyl ether) (PFA), ethylene vinyl alcohol, polyolefin, polystyrene, epoxy, polyvinylidene chloride (PVdC), poly(styrene-b-isobutylene-b-styrene) (for example, SIBS and/or SIBS 50A), polycarbonates, ionomers, biocompatible polymers, other suitable materials, or mixtures, combinations, copolymers thereof, polymer/metal composites, and the like. In some embodiments the sheath can be blended with a liquid crystal polymer (LCP).

(71) Some examples of suitable metals and metal alloys include stainless steel, such as 304V, 304L, and 316LV stainless steel; mild steel; nickel-titanium alloy such as linear-elastic and/or super-elastic nitinol; other nickel alloys such as nickel-chromium-molybdenum alloys (e.g., UNS: N06625 such as INCONEL® 625, UNS: N06022 such as HASTELLOY® C-22®, UNS: N10276 such as HASTELLOY® C276®, other HASTELLOY® alloys, and the like), nickel-copper alloys (e.g., UNS: N04400 such as MONEL® 400, NICKELVAC® 400, NICORROS® 400, and the like), nickel-cobalt-chromium-molybdenum alloys (e.g., UNS: R30035 such as MP35-N® and the like), nickel-molybdenum alloys (e.g., UNS: N10665 such as HASTELLOY® ALLOY B2®), other nickel-chromium alloys, other nickel-molybdenum alloys, other nickel-cobalt alloys, other nickel-iron alloys, other nickel-copper alloys, other nickel-tungsten or tungsten alloys, and the like; cobalt-chromium alloys; cobalt-chromium-molybdenum alloys (e.g., UNS: R30003 such as ELGILOY®, PHYNOX®, and the like); platinum enriched stainless steel; titanium; combinations

thereof; and the like; or any other suitable material.

(72) In at least some embodiments, portions or all of the medical device system **10** and components thereof may also be doped with, made of, or otherwise include a radiopaque material. Radiopaque materials are understood to be materials capable of producing a relatively bright image on a fluoroscopy screen or another imaging technique during a medical procedure. This relatively bright image aids the user of the shaft in determining its location. Some examples of radiopaque materials can include, but are not limited to, gold, platinum, palladium, tantalum, tungsten alloy, polymer material loaded with a radiopaque filler, and the like. Additionally, other radiopaque marker bands and/or coils may also be incorporated into the design of the medical device system **10** and components thereof to achieve the same result.

(73) In some embodiments, a degree of Magnetic Resonance Imaging (MRI) compatibility is imparted into the shaft. For example, the medical device system **10** and components thereof may include a material that does not substantially distort the image and create substantial artifacts (e.g., gaps in the image). Certain ferromagnetic materials, for example, may not be suitable because they may create artifacts in an MRI image. The medical device system **10** and components thereof may also be made from a material that the MRI machine can image. Some materials that exhibit these characteristics include, for example, tungsten, cobalt-chromium-molybdenum alloys (e.g., UNS: R30003 such as ELGILOY®, PHYNEX®, and the like), nickel-cobalt-chromium-molybdenum alloys (e.g., UNS: R30035 such as MP35-N® and the like), nitinol, and the like, and others.

(74) It should be understood that this disclosure is, in many respects, only illustrative. Changes may be made in details, particularly in matters of shape, size, and arrangement of steps without exceeding the scope of the disclosure. This may include, to the extent that it is appropriate, the use of any of the features of one example embodiment being used in other embodiments. The disclosure's scope is, of course, defined in the language in which the appended claims are expressed.

Claims

1. A system for delivering an implantable heart valve, comprising: a connection assembly having an engaged configuration and a disengaged configuration, the connection assembly including: an inner shaft having a proximal end region, a distal end region and a first coupling member disposed along a portion of the distal end region, wherein the first coupling member includes a plurality of apertures; a support shaft having a proximal end region, a distal end region and a second coupling member disposed along a portion of the proximal end region, wherein the second coupling member includes a stem, wherein the stem includes a groove extending circumferentially around the stem; a plurality of bearings, wherein each bearing is disposed in a respective aperture of the plurality of apertures, and wherein a portion of each bearing is also disposed in a portion of the groove when the connection assembly is in the engaged configuration; and a locking cap coupled to the inner shaft, wherein a portion of the locking cap is configured to cover at least a portion of each of the plurality of bearings when the connection assembly is in the engaged configuration.
2. The system of claim 1, wherein the locking cap includes a plurality of tabs extending circumferentially around a longitudinal axis of the locking cap.
3. The system of claim 2, wherein each of the plurality of tabs is configured to flex radially away from the longitudinal axis.
4. The system of claim 3, wherein each of the plurality of tabs is configured to flex radially away from the longitudinal axis as the stem is inserted into a lumen of the first coupling member.
5. The system of claim 4, wherein each of the plurality of tabs is configured to exert a radially inward force on each of the plurality of bearings in the engaged configuration.
6. The system of claim 5, wherein each of the plurality of bearings is configured to slide circumferentially around the stem while disposed within the groove.

7. The system of claim 6, wherein each of the bearings is configured to rotate relative to the first coupling member, the second coupling member and the locking cap in the engaged configuration.
 8. The system of claim 7, wherein each of the bearings is configured to shift radially outward within its respective aperture of the plurality of apertures as the stem is inserted into a lumen of the first coupling member.
 9. The system of claim 8, wherein each of the plurality of bearings is disposed within its respective aperture of the plurality of apertures in the disengaged configuration.
 10. The system of claim 1, wherein the locking cap is fixedly attached to the first coupling member.
 11. The system of claim 10, wherein the first coupling member is configured to rotate relative to the second coupling member in the engaged configuration.
 12. A system for delivering an implantable heart valve, comprising: a tip assembly having a distal end region and a proximal end region; a guidewire shaft coupled to the distal end region of the tip assembly; a connection assembly having an engaged configuration and a disengaged configuration, the connection assembly including: an inner shaft having a proximal end region, a distal end region and a first coupling member disposed along a portion of the distal end region, wherein the first coupling member includes a plurality of apertures; a support shaft having a proximal end region, a distal end region and a second coupling member disposed along a portion of the proximal end region, wherein the second coupling member includes a stem, wherein the stem includes a groove extending circumferentially around the stem; a plurality of bearings, wherein each bearing is disposed in a respective aperture of the plurality of apertures, and wherein a portion of each bearing is also disposed in a portion of the groove when the connection assembly is in the engaged configuration; and a locking cap coupled to the inner shaft, wherein a portion of the locking cap is configured to cover at least a portion of each of the plurality of bearings when the connection assembly is in the engaged configuration.
 13. The system of claim 12, wherein the locking cap includes a plurality of tabs extending circumferentially around a longitudinal axis of the locking cap.
 14. The system of claim 13, wherein each of the plurality of tabs is configured to flex radially away from the longitudinal axis.
 15. The system of claim 14, wherein each of the plurality of tabs is configured to flex radially away from the longitudinal axis as the stem is inserted into a lumen of the first coupling member.
 16. The system of claim 15, wherein each of the plurality of tabs is configured to exert a radially inward force on each of the plurality of bearings in the engaged configuration.
 17. The system of claim 16, wherein each of the plurality of bearings is configured to slide circumferentially around the stem while disposed within the groove.
 18. The system of claim 17, wherein each of the bearings is configured to rotate relative to the first coupling member, the second coupling member and the locking cap in the engaged configuration.
 19. The system of claim 18, wherein the first coupling member is configured to rotate relative to the second coupling member in the engaged configuration.
 20. A method for delivering an implantable heart valve, the method comprising: attaching a first coupling member of an actuation shaft to a second coupling member of a support shaft of a medical device delivery system, the medical device delivery system including the implantable heart valve; wherein attaching the first coupling member of the actuation shaft to the second coupling member of the support shaft includes positioning a plurality of bearings into a groove of the second coupling member; advancing the medical device delivery system to a target site adjacent the heart; and deploying the implantable heart valve at the target site.
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